

Deterministic Optimization

Large-Scale Optimization

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Primal-Dual Relationship: Pricing
Problem and Separation Problem

Large-Scale Optimization

Learning Objectives for this lesson

- Further explore the primal-dual relationship between column generation and constraint generation
- Relate pricing problem in column generation to separation problem in constraint generation

Constraint Generation

- To solve an optimization problem with many constraints, we can use constraint generation:
 1. Start with a reduced master problem with a subset of constraints
 2. Solve the reduced master problem with optimal solution x^*
 3. Check if all the constraints (especially those not in the reduced master problem) are satisfied at x^*
 - a. If yes, then the current solution x^* is optimal for master problem
 - b. Otherwise, find a constraint violated by x^* , add it to reduced master problem. Go to Step 2.

Constraint Generation & Separation

$$\begin{aligned} (DMP) \quad Z_D = \max \quad & \sum_{i=1}^m b_i y_i \\ \text{s.t.} \quad & \mathbf{y}^\top \mathbf{A}_j \leq c_j, \quad \forall j = 1, \dots, N \end{aligned}$$

$$\begin{aligned} (RDMP) \quad Z_{RD} = \max \quad & \sum_{i=1}^m b_i y_i \\ \text{s.t.} \quad & \mathbf{y}^\top \mathbf{A}_j \leq c_j, \quad \forall j \in I. \end{aligned}$$

$\mathbf{y}^*$: = optimum
of (RDMP)

Separation Problem

$\hat{Z} = \min_{j=1, \dots, N} \{c_j - (\mathbf{y}^*)^\top \mathbf{A}_j\}$ (constraint violation)
Let A_j be the constraint achieving the minimum.

$\hat{Z} < 0$?

No

Stop

Yes

Add A_j to I

Return new I

Column Generation & Pricing

$$\begin{aligned}
 (MP) \quad Z_P = \min \quad & \sum_{j=1}^N c_j x_j \\
 \text{s.t.} \quad & \sum_{j=1}^N a_{ij} x_j = b_i, \quad \forall i = 1, \dots, m \\
 & \text{(Equivalently: } \sum_{j=1}^N \mathbf{A}_j x_j = \mathbf{b} \text{)} \\
 & x_j \geq 0, \quad \forall j = 1, \dots, N,
 \end{aligned}$$

$$\begin{aligned}
 (RMP) Z_R = \min \quad & \sum_{j \in I} c_j x_j \\
 \text{s.t.} \quad & \sum_{j \in I} a_{ij} x_j = b_i, \quad \forall i = 1, \dots, m \\
 & \text{(Equivalently: } \sum_{j \in I} \mathbf{A}_j x_j = \mathbf{b} \text{)} \\
 & x_j \geq 0, \quad \forall j \in I.
 \end{aligned}$$

$$(\mathbf{y}^*)^\top = \mathbf{c}_B^\top \mathbf{B}^{-1}$$

Pricing Problem

$$\hat{Z} = \min_{j=1, \dots, N} \{c_j - (\mathbf{y}^*)^\top \mathbf{A}_j\} \text{ (reduced cost)}$$

Let $\mathbf{A}_j$ be the column achieving the minimum.

$$\hat{Z} < 0?$$

No

Stop

Yes

Add $\mathbf{A}_j$ to I

Return new I

Primal Pricing = Dual Separation

- There is again a complete symmetry:
 - Pricing problem in column generation finds the minimum reduced cost of all primal variables
 - Separation problem in constraint generation finds the largest constraint violation in dual constraints
 - These two problems are the same problem viewed in primal and dual perspective

Implication for Pricing Problem in Column Generation

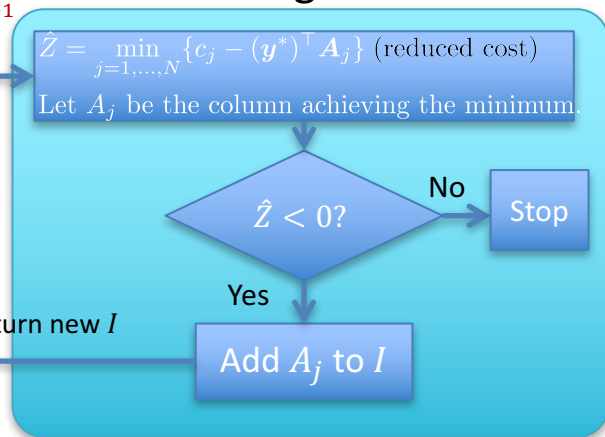
- An implication is that the y^* can take any dual feasible solution
 - When Simplex method solve (RMP), it produces a dual solution
 - You don't have to invert B to get y^*

$$(y^*)^T = c_B^T B^{-1}$$

Pricing Problem

$$\begin{aligned}
 (MP) \quad Z_P = \min \quad & \sum_{j=1}^N c_j x_j \\
 \text{s.t.} \quad & \sum_{j=1}^N a_{ij} x_j = b_i, \quad \forall i = 1, \dots, m \\
 & \text{(Equivalently: } \sum_{j=1}^N A_j x_j = b) \\
 & x_j \geq 0, \quad \forall j = 1, \dots, N,
 \end{aligned}$$

$$\begin{aligned}
 (RMP) Z_R = \min \quad & \sum_{j \in I} c_j x_j \\
 \text{s.t.} \quad & \sum_{j \in I} a_{ij} x_j = b_i, \quad \forall i = 1, \dots, m \\
 & \text{(Equivalently: } \sum_{j \in I} A_j x_j = b) \\
 & x_j \geq 0, \quad \forall j \in I.
 \end{aligned}$$



Summary

- We learned:
 - Yet another key relationship between column generation and constraint generation, namely pricing problem and separation problem are the same problem viewed from primal-dual perspective